about this book

The primary goal of this book is to empower web developers to build dynamic and intelligent websites and applications that use the full potential of large language models and other generative AI capabilities. I aim to equip you with the practical tools and techniques necessary to integrate sophisticated AI features—such as retrieval-augmented generation (RAG), document summarization, and interactive chatbots—directly into your web-based projects, all within the familiar JavaScript ecosystem. You'll also get a tutorial on interacting with the Model Context Protocol to seamlessly extend capabilities beyond what we build ourselves.

Who should read this book

This book is tailored for frontend developers and full-stack web developers who are comfortable with React and Next.js. If you have a basic proficiency in these technologies and are looking to incorporate cutting-edge AI functionalities into your web applications, this book is for you. No prior experience with Python or machine learning is required; we focus entirely on JavaScript and other tools common to web developers. A basic awareness of generative AI concepts and prior experience using tools like ChatGPT or similar interfaces will be beneficial, but the foundational chapters will get you up to speed.

How this book is organized: A road map

This book is divided into 4 parts spanning 12 chapters, designed to progressively build your expertise in full-stack generative AI application development.

- Part 1: Building basic generative AI web apps (chapters 1–4) lays the essential groundwork. I begin by defining what generative AI web applications are, their